

Owed lemmas for the rank-two atomic residue proof

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Purpose. The argument transcribed in `2026-08-15-c912-atom-residue-proof.tex` uses, at its first and most load-bearing step, horizontality of the Poincaré pairing for the A -model connection. Its independent verification (`2026-08-15-c912-atom-residue-proof-verification.md`) found that this is *not* an available import: Katzarkov–Kontsevich–Pantev–Yu state horizontality only after assuming convergence and passing to the complex-analytic bundle, whereas the A -model F -bundle of a projective variety is non-archimedean analytic. The step is therefore ours to prove. This note proves it, over the Novikov ring and with no convergence hypothesis, and lists what else that verification recorded as owed.

1 Setting

Let T be a smooth projective variety over $\mathbf{C}$ of dimension n , and let Λ be the completed Novikov ring of its numerical curve lattice. All statements below are identities in Λ -modules; no convergence of any Gromov–Witten generating function is used, and no base change to a complex-analytic bundle is performed.

Write $H = H^{\text{ev}}(T) \otimes \Lambda$ for the even part, $(\cdot, \cdot)$ for the Poincaré pairing extended Λ -bilinearly, $\star_\tau$ for the genus-zero quantum product at a bulk parameter τ , and

$$\text{Gr}(\phi) = \frac{1}{2}(\deg \phi - n)\phi \quad \text{for homogeneous } \phi,$$

for the grading operator. The connection is

$$\nabla_u \partial_u = u \partial_u - u^{-1} \mathcal{E} \star_\tau + \text{Gr}, \quad \nabla_\xi = \xi + u^{-1} C_\xi, \quad C_\xi = \xi \star_\tau, \quad (1)$$

so that horizontal sections satisfy

$$u \partial_u y = A(u)y, \quad A(u) = u^{-1}U - \text{Gr}, \quad U := \mathcal{E} \star_\tau; \quad \xi y = B_\xi(u)y, \quad B_\xi(u) = -u^{-1}C_\xi. \quad (2)$$

The pairing is taken in the usual twisted form: for sections $y, \tilde{y}$ one considers $(y(u), \tilde{y}(-u))$.

2 The two adjointness identities

Lemma 2.1 (Frobenius, over Λ). *For all $a, b, c \in H$ and every bulk parameter τ ,*

$$(a \star_\tau b, c) = (a, b \star_\tau c).$$

In particular every quantum multiplication operator, including $U = \mathcal{E} \star_\tau$ and each $C_\xi = \xi \star_\tau$, is self-adjoint for the Poincaré pairing.

Proof. By definition the structure constants of $\star_\tau$ are

$$(a \star_\tau b, c) = \sum_d \sum_{m \geq 0} \frac{Q^d}{m!} \langle a, b, c, \tau, \dots, \tau \rangle_{0, m+3, d},$$

a formal sum over effective numerical classes d with coefficients the genus-zero Gromov–Witten invariants. Those invariants are defined by integration against the virtual class of a moduli space of stable maps with *unordered* marked-point labels up to the specified insertions, hence are symmetric under permuting their arguments; on the even part no Koszul signs arise. The displayed expression is therefore totally symmetric in a, b, c , and comparing it with the same expression for $(a, b \star_\tau c)$ gives the claim.

Two remarks on what this does and does not use. The identity is an identity of formal series: it holds coefficientwise in Q^d and in the bulk parameter, so it is a statement in Λ and requires no convergence. And it is exactly the total symmetry of the third derivatives of the genus-zero potential, so it is the Frobenius property in its usual form, not a strengthening of it. $\square$

Lemma 2.2 (The grading operator is anti-self-adjoint). *For all $a, b \in H$, $(\text{Gr}a, b) + (a, \text{Gr}b) = 0$.*

Proof. Both sides are Λ -bilinear, so it suffices to check on homogeneous classes. The Poincaré pairing of $\phi \in H^k$ with $\psi \in H^\ell$ vanishes unless $k + \ell = 2n$. In that case

$$(\text{Gr}\phi, \psi) + (\phi, \text{Gr}\psi) = \left[\frac{1}{2}(k - n) + \frac{1}{2}(\ell - n) \right] (\phi, \psi) = \frac{1}{2}[(k + \ell) - 2n] (\phi, \psi) = 0.$$

This is a statement about the cohomological grading and Poincaré duality alone; the Novikov variables play no role, and Gr is Λ -linear because it acts on the coefficient ring trivially. $\square$

Remark 2.3. Lemma 2.2 is where the normalization of Gr by $-n/2$ earns its keep: with any other additive constant the operator would fail to be anti-self-adjoint, and horizontality below would acquire a scalar defect.

3 Horizontality

Theorem 3.1 (The Poincaré pairing is horizontal, over Λ). *Let $y, \tilde{y}$ be horizontal sections of (1). Then*

$$(y(u), \tilde{y}(-u))$$

is constant in u and in every base direction. Equivalently, with P_0 the Gram matrix of the Poincaré pairing in a fixed basis,

$$A(u)^T P_0 + P_0 A(-u) = 0, \quad B_\xi(u)^T P_0 + P_0 B_\xi(-u) = 0. \quad (3)$$

Proof. Put $f(u) = y(u)^T P_0 \tilde{y}(-u)$. Substituting $w = -u$ shows $u\partial_u$ is invariant under $u \mapsto -u$, so $(u\partial_u)(\tilde{y}(-u)) = A(-u)\tilde{y}(-u)$, and therefore

$$u\partial_u f = y^T [A(u)^T P_0 + P_0 A(-u)] \tilde{y}(-u).$$

Now compute the bracket from (2):

$$(u^{-1}U - \text{Gr})^T P_0 + P_0(-u^{-1}U - \text{Gr}) = u^{-1}(U^T P_0 - P_0 U) - (\text{Gr}^T P_0 + P_0 \text{Gr}).$$

The first bracket vanishes by Lemma 2.1, since U is a quantum multiplication operator, and the second by Lemma 2.2. Hence $u\partial_u f = 0$.

For a base direction, $B_\xi(u) = -u^{-1}C_\xi$ has no Gr term and is odd in u , so

$$B_\xi(u)^T P_0 + P_0 B_\xi(-u) = -u^{-1}(C_\xi^T P_0 - P_0 C_\xi) = 0$$

by Lemma 2.1 again, applied to $C_\xi = \xi \star_\tau$. So f is constant in every base direction as well. $\square$

Corollary 3.2 (What the atomic residue proof needs). *In the notation of 2026-08-15-c912-atom-residue-p equation (7) holds with the constant Gram matrix P_0 , and hence so do its consequences (8), (10) and (11).*

Proof. Equation (7) there is $u\partial_u P(u) + A(u)^T P(u) + P(u)A(-u) = 0$ for a possibly u -dependent pairing $P(u)$. Theorem 3.1 gives it with $P(u) = P_0$ constant, so the $u\partial_u P$ term drops and the remaining identity is (3). The u^{-1} coefficient is $N^T P_0 = P_0 N$, which is (8); the u^0 coefficient with a constant pairing is $A_0^T P_0 + P_0 A_0 = 0$, which is (10) with $P_1 = 0$, and the sandwich argument producing $A_0 L \subset L$ goes through unchanged — indeed more simply, since the terms of (10) involving N and P_1 are absent. $\square$

Remark 3.3 (What was actually at issue). The verification's concern was not that horizontality is false but that the source establishes it only after assuming convergence and passing to the complex-analytic bundle, while the object the proof works with is non-archimedean. Theorem 3.1 removes that dependence: both inputs are algebraic identities in Λ -modules, one the total symmetry of genus-zero Gromov–Witten invariants and the other a statement about the cohomological grading. Neither refers to a topology on Λ . In particular nothing here touches the Euler-pairing comparison or the Serre automorphism, which are the objects the source does defer.

4 Still owed

The same verification recorded four further unstated inputs. They are bookkeeping rather than substance, but none is written:

- (i) that the even part is a sub- F -bundle, so that restricting the whole construction to it is legitimate;
- (ii) uniqueness and gluing of the spectral splitting;
- (iii) the identity principle on irreducible k -analytic spaces, which is what the crossing-point argument uses when it propagates constancy from a nonempty open subset;
- (iv) descent of the invariant through the blowup, projective-bundle and disjoint-union equivalences, and not only through the two equivalences in the source's own definition.

Separately, and not bookkeeping: the parity ranks $(2, 10)$ rest on the source's assertion that the zero-eigenvalue packet does not split further at the relevant point, whose cited support carries no proof and covers only a restricted case. That is the one place where the argument is genuinely conditional, and it is a finite splitting statement about a single packet rather than an invariance statement about arbitrary centres.